

## Team Minutes for TERMINATOR :)

| Team Members' Names |
|---------------------|
| Jessica Batson      |
| Kyle Ha             |
| Lydia Tan           |
| Julia Bickley       |

### Meeting 1

|                                               |                                                                                                                                                                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Meeting Date:</b><br><b>02/11/2026</b>     | <b>Start Time: 8:00am</b><br><b>End Time: 9:03am</b><br><b>Summary of Meeting</b>                                                                                                                                      |
| 1. Attendance                                 | <i>Record members present (if absent, include reasons and whether teammates were notified in advance).</i><br>- Kyle Ha<br>- Jessica Batson<br>- Lydia Tan<br>- Julia Bickley                                          |
| 2. Purpose of Meeting                         | <i>Reviewing assignment/task of the Mini Project</i><br>- Goal: Delegate tasks for each member and decide the schedule until the Mini Project due date (2/23).                                                         |
| 3. Discuss work completed since last meeting. | N/A<br>                                                                                                                            |
| 4. Summarize work completed during meeting    | <i>Summarize discussion and work accomplished.</i><br><br><i>Discussion:</i><br><i>Decide the tasks for each member, the deliverables of the Mini Project, and discuss schedules for everyone until 2/23, since it</i> |

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                      | <p><i>is Midterm Time.</i></p> <p><i>Work-Accomplished:</i><br/> <i>Finished Team Charter + Currently further reviewing the details of the Mini Project Document.</i></p>                                                                                                                                                                                                                                                                                                 |
| 5. Review action items to be completed after meeting | <p><i>Who will do what by when?</i></p> <ul style="list-style-type: none"> <li>- Kyle Ha:<br/> <i>Assembly of robot. Implementing PI controller / code.</i></li> <li>- Jessica Batson:<br/> <i>Display an updated pic of the camera &amp; indicate position of the marker</i></li> <li>- Lydia Tan:<br/> <i>Add position control onto Simulink block diagram. Assembly of robot.</i></li> <li>- Julia Bickley:<br/> <i>Display goal position on LCD screen</i></li> </ul> |
| 6. Schedule next meeting                             | <p><i>2/18 at 8-9:50 (Lab Time), checkup + testing.</i><br/> <i>*Meet during lecture on 2/18</i></p>                                                                                                                                                                                                                                                                                                                                                                      |
| 7. Recording secretary                               | <p><i>Jessica Batson – 2/11/2026</i></p>                                                                                                                                                                                                                                                                                                                                                                                                                                  |

## Meeting 2

|                                                      |                                                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Meeting Date:</b><br><b>2/18/2026</b>             | <b>Start Time: 9:00 a.m.</b><br><b>End Time: 9:20 a.m.</b><br><b>Summary of Meeting</b>                                                                                                                                                                                                                                      |
| 1. Attendance                                        | Record members present (if absent, include reasons and whether teammates were notified in advance).<br>- Kyle Ha<br>- Jessica Batson<br>- Lydia Tan<br>- Julia Bickley                                                                                                                                                       |
| 2. Purpose of Meeting                                | Determine what's left to do to finish up the Mini Project, which includes putting together the Controls and CV part with each other and testing if the robot accurately moves its position based on the Pi's input. We also need to finish up the Simulink plotting and determine the best Kp values to match actual values. |
| 3. Discuss work completed since last meeting.        | Team member: Lydia Tan<br>Work Completed: Finished making the Simulink for the motor control for the robot.                                                                                                                                                                                                                  |
|                                                      | Team member: Kyle Ha<br>Work Completed: Wrote the position code to add onto the Assignment 2 Code to also be able to parse the Pi values into the Arduino Code.                                                                                                                                                              |
|                                                      | Team member: Jessica Batson<br>Work Completed: Finished writing the code to make sure the camera is able to find the marker and determine the direction the marker is facing in which quadrant.                                                                                                                              |
|                                                      | Team member: Julia Bickley<br>Work Completed: Finished writing the code to output the values from Jessica's code to the LCD Monitor.                                                                                                                                                                                         |
|                                                      |                                                                                                                                                                                                                                                                                                                              |
| 4. Summarize work completed during meeting           | We were able to finally put together the Pi and Arduino codes and set up the camera onto the robot. We were able to troubleshoot why the Arduino Code was not working properly and forcing the robot to not move even though the camera was detecting different directions.                                                  |
| 5. Review action items to be completed after meeting | Team member: Lydia Tan<br>Need to finish the MATLAB code for plotting the Simulink graph vs. Actual values from the robot. Assist Kyle in troubleshooting the position of Arduino code.                                                                                                                                      |
|                                                      | Team member: Kyle Ha<br>Work on troubleshooting the position of Arduino code and test if it finally works with the Pi.                                                                                                                                                                                                       |
|                                                      | Team member: Jessica Batson<br>Done Yippee :>                                                                                                                                                                                                                                                                                |
|                                                      | Team member: Julia Bickley<br>Done Yippee: >                                                                                                                                                                                                                                                                                 |

|                          |                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------|
|                          | <i>Just need to attach the LCD monitor to the robot = team effort</i>                                     |
|                          | <i>Team member:</i><br> |
| 6. Schedule next meeting | <i>February 25<sup>th</sup>, 2026, During Lab Time</i>                                                    |
| 7. Recording secretary   | <i>Lydia Tan – 2/18/2026</i>                                                                              |

### **Demo 1 Meeting 1**

|                      |                              |
|----------------------|------------------------------|
| <b>Meeting Date:</b> | <b>Start Time: 9:00 a.m.</b> |
|----------------------|------------------------------|

|                                                      |                                                                                                                                                                                                                                                                               |
|------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>2/27/2026</b>                                     | <b>End Time: 9:30 a.m.</b><br><b>Summary of Meeting</b>                                                                                                                                                                                                                       |
| 1. Attendance                                        | <ul style="list-style-type: none"> <li>- Kyle Ha</li> <li>- Jessica Batson</li> <li>- Lydia Tan</li> <li>- Julia Bickley</li> </ul>                                                                                                                                           |
| 2. Purpose of Meeting                                | <i>Double check everything is going well with the robot with a new motor shield. Go over Demo 1 Assignment so we can begin to set roles within steering control and camera vs real world.</i>                                                                                 |
| 3. Discuss work completed since last meeting.        | <i>Team member: Lydia Tan</i><br><i>Mini Project: Debugged the Integral Controller code and fixed the I2C Connection part of the code for Arduino, to make it compatible with the Raspberry Pi. Contact Jo and Sharma regarding the Technical Review of the Mini Project.</i> |
|                                                      | <i>Team member: Kyle Ha</i><br><i>Mini Project</i>                                                                                                                                                                                                                            |
|                                                      | <i>Team member: Jessica Batson</i><br><i>Mini Project</i>                                                                                                                                                                                                                     |
|                                                      | <i>Team member: Julia Bickley</i><br><i>Mini Project</i>                                                                                                                                                                                                                      |
|                                                      |                                                                                                                                                                                                                                                                               |
| 4. Summarize work completed during meeting           | <i>Did the Technical Review with Sharma + Got a new motor shield, rewired and retested Mini Project. We went over Demo 1 assignment so we can determine what needs to be done before/what needs to be started before the next meeting and lab.</i>                            |
| 5. Review action items to be completed after meeting | <i>Team member: Lydia Tan</i><br>Ensure the proper implementation of the control system by going through each step of the steering control handout and finding correct encoder sign conventions as well as calculating forward and rotational velocity.                       |
|                                                      | <i>Team member: Kyle Ha</i><br>Ensure the proper implementation of the control system by going through each step of the steering control handout and finding correct encoder sign conventions as well as calculating forward and rotational velocity.                         |
|                                                      | <i>Team member: Jessica Batson</i><br><i>Start working on camera calibration for CV</i>                                                                                                                                                                                       |
|                                                      | <i>Team member: Julia Bickley</i>                                                                                                                                                                                                                                             |

|                          |                                                                                    |  |
|--------------------------|------------------------------------------------------------------------------------|--|
|                          |  |  |
| 6. Schedule next meeting | <i>February 4<sup>th</sup>, 2026, During Lab Time</i>                              |  |
| 7. Recording secretary   | <i>Kyle Ha – 2/27/2026</i>                                                         |  |

## Demo 1 Meeting 2

|                                               |                                                                                                                                    |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Meeting Date:</b><br><b>3/6/2026</b>       | <b>Start Time: 9:30 a.m.</b><br><b>End Time: 10:00 a.m.</b><br><b>Summary of Meeting</b>                                           |
| 1. Attendance                                 | Lydia Tan<br>Julia Bickley                                                                                                         |
| 2. Purpose of Meeting                         | <i>Discuss progress in Demo 1 codes and what to do from here looking towards Demoing.</i>                                          |
| 3. Discuss work completed since last meeting. | Team member: Lydia Tan<br>- Wrote the distance part of the LC Code. (Still need to be calibrated)                                  |
|                                               | Team member: Kyle Ha<br>- Wrote the rotation part of the LC Code. (Still need to be calibrated)                                    |
|                                               | Team member: Julia Bickley<br>- Collaborated with jessica to write distortion of the camera and angle detection code (in progress) |
|                                               | Team member: Jessica Batson<br>- Worked on the distortion of the camera                                                            |
|                                               | Team member:<br>                               |

|                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>4. Summarize work completed during meeting</p>           | <p>LC:</p> <ul style="list-style-type: none"> <li>- Lydia: Worked on the Distance Code, was able to test out the robot using a very long 10 feet piece of tape on the floor, and calibrating the distance the robot travels. Currently testing it on carpet floor, have not tested out yet for different surfaces.</li> <li>- Kyle: Sent the rotation side of the code to Lydia.</li> </ul> <p>CV:</p> <ul style="list-style-type: none"> <li>- Julia: worked on camera angle detection code and collaborated with Jessica.</li> </ul> |
| <p>5. Review action items to be completed after meeting</p> | <p>Team member: Lydia Tan</p> <ul style="list-style-type: none"> <li>- Test the robot on different surfaces (smooth + rough) to see if the PI controller is working and it's traveling the correct amount of distance. Will also work to fuse together the distance and rotation code.</li> <li>- Make sure everything is ready for demoing</li> </ul>                                                                                                                                                                                 |
|                                                             | <p>Team member: Kyle Ha</p> <ul style="list-style-type: none"> <li>- Fix the Rotation Code so that the right wheel works correctly (Left wheel works). Fine tune the entire combined code after Lydia sends it.</li> <li>- Make sure everything is ready for demoing</li> </ul>                                                                                                                                                                                                                                                        |
|                                                             | <p>Team member: Jessica Batson</p> <ul style="list-style-type: none"> <li>- Make sure everything is ready for demoing</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                             | <p>Team member: Julia Bickley</p> <ul style="list-style-type: none"> <li>- Fix angle detection pi code, completed CV side of coding.</li> <li>- Work with Lydia to fuse codes</li> <li>- Make sure everything is ready for demoing</li> </ul>                                                                                                                                                                                                                                                                                          |
|                                                             | <p>Team member:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |



|                          |                                                                                     |
|--------------------------|-------------------------------------------------------------------------------------|
|                          |  |
| 6. Schedule next meeting | <i>March 13<sup>th</sup>, 2026 – During Lab Time</i>                                |
| 7. Recording secretary   | <i>Julia Bickley – 3/6/2026</i>                                                     |

## Demo 2 Meeting 1

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Meeting Date:</b><br><b>3/16/2026</b>             | <b>Start Time: 8am End Time: 10am</b><br><b>Summary of Meeting</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 1. Attendance                                        | Jessica Batson<br>Julia Bickley<br>Kyle Ha<br>Lydia Tan                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| 2. Purpose of Meeting                                | <i>The purpose of this meeting is to begin working on Demo2 and split up tasks on both the LC and CV subsystems.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 3. Discuss work completed since last meeting.        | <i>Jessica Batson: Began working on the CV PI logic and implemented color detection to the Demo1 code.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                      | <i>Julia Bickley: began developing CV Pi logic and adding color detection + logic to Demo1 code.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                      | <i>Kyle Ha: Began brainstorming how to integrate CV Pi logic into the Controls side of detection and alignment using the information given from the camera.</i>                                                                                                                                                                                                                                                                                                                                                                                            |
|                                                      | <i>Lydia Tan: Listed out the to-do List for LC and created the subsystems for LC members to work on. Started working on the MATLAB Simulink pulled over from Demo 1 and started applying it to Demo 2.</i>                                                                                                                                                                                                                                                                                                                                                 |
| 4. Summarize work completed during meeting           | <i>The work completed today included most of the CV logic code so that the code can be given to LC to further implement the Arduino code. For the LC side, finished building the area on the robot to hold the Pi and the Power bank. Meanwhile, the LC side started creating the base code for the subsystems in preparation for the completed CV code.<br/>Tons of brainstorming on the controls side on detection, alignment, distance, turning, etc as we wait for CV logic code to be finished and can be fully integrated into the Arduino code.</i> |
| 5. Review action items to be completed after meeting | <i>Jessica Batson: Fix color detection red so that it will constantly show red detection rather than jumping from red to none constantly.</i>                                                                                                                                                                                                                                                                                                                                                                                                              |
|                                                      | <i>Julia Bickley: improve color detection for red and green and ensure the detected color is consistent</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|                                                      | <i>Kyle Ha: Following the CV side, begin to start a layout of a working detection and alignment code working alongside Pi code.</i>                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                                                      | <i>Lydia Tan: Finish the Distance and Turn 90 degrees Code and continue creating the MATLAB Simulink.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 6. Schedule next meeting                             | 3/30/2026                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 7. Recording secretary                               | Jessica Batson                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## Demo 2 Meeting 2

|                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Meeting Date:</b><br><b>3/30/2026</b>             | <b>Start Time: 8am End Time: 10am</b><br><b>Summary of Meeting</b>                                                                                                                                                                                                                                                                                                                                                                                     |
| 1. Attendance                                        | Jessica Batson<br>Julia Bickley<br>Kyle Ha<br>Lydia Tan                                                                                                                                                                                                                                                                                                                                                                                                |
| 2. Purpose of Meeting                                | Implementing the PI code and Arduino code to accurately move the robot.                                                                                                                                                                                                                                                                                                                                                                                |
| 3. Discuss work completed since last meeting.        | Jessica Batson: changed the PI code so that the Pi can be run continuously without a monitor being plugged in.                                                                                                                                                                                                                                                                                                                                         |
|                                                      | Julia Bickley: changed the Pi code's color detection and integration with arduino so it runs without being plugged into the monitor.                                                                                                                                                                                                                                                                                                                   |
|                                                      | Kyle Ha: Continued completing the alignment and detection code so it will function alongside the CV's working Pi code and attach more building components to fit for the addition of the monitor and power supply.                                                                                                                                                                                                                                     |
|                                                      | Lydia Tan: Worked on combining all the subsystems together: Search, Align, Distance, then Turn. Mainly calibration and debugging since there was a lot of disturbance between the Camera and the Pi's Outputs.                                                                                                                                                                                                                                         |
| 4. Summarize work completed during meeting           | We have finished independently writing code on both the LC and CV side and now need to implement both sides together. This involved insuring I2C communication was used. On the LC side, finished creating the individual subsystem codes, mainly worked on debugging certain issues in the overall full code and testing the robot at different angles and situations and performing lots of trials and errors to try and increase the robot's speed. |
| 5. Review action items to be completed after meeting | Jessica Batson: Fix the camera (there are some issues with changing this code that when the PI is plugged into the monitor it no longer shows the camera when run)                                                                                                                                                                                                                                                                                     |
|                                                      | Julia Bickley: Fix the camera (above) and ensure Arduino and Pi are communicating properly.                                                                                                                                                                                                                                                                                                                                                            |
|                                                      | Kyle Ha: Have a working code compatible to integrate with other portions of controls as well as following the guidelines of the Pi.                                                                                                                                                                                                                                                                                                                    |
|                                                      | Lydia: Finish testing the robot and fix any issues with the code so that the robot can perform the demo.                                                                                                                                                                                                                                                                                                                                               |
| 6. Schedule next meeting                             | After Demo 2                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 7. Recording                                         | Lydia Tan                                                                                                                                                                                                                                                                                                                                                                                                                                              |

|           |  |
|-----------|--|
| secretary |  |
|-----------|--|